

2.2. Deductive Method

Deductive method is the opposite of the inductive method. It is based on deduction. In this method we proceed from general to particular and from abstract to concrete. In this method the rules are given at the very outset. The students are asked to apply these rules to solve more problems. In this method the students accept the formula as pre-established and well-established truth, e.g. the formula.

$$\text{Simple Interest} = \frac{\text{Principal} \times \text{rate} \times \text{time}}{100}$$

can be used by the students to solve problems.

a) Merits of Deductive Method

- i. It is a short and time saving method.
- ii. This method suits all types of students.
- iii. This method is suitable for all topics.
- iv. This method provides sufficient practice in the application of various mathematical formulae and rules.
- v. The speed and efficiency increases by use of this method.
- vi. It completes the process of inductive. Probability in induction gets converted into certainty by this method.

b) Demerits of Deductive Method

- i. It is not a psychological method.
- ii. It encourages cramming.
- iii. It taxes mind of students.
- iv. Students get disinterested, as they are only passive listeners.
- v. It puts more emphasis on memory.
- vi. It is not found quite suitable for the development of thinking, reasoning and discovery.

c) Place of Deductive Method

It is a method of work and practice. While applying deduction, great care should be exercised so that no wrong creeps in. This method works well with average students and teachers. This method provides skill, speed and efficiency in solving problems. Most teachers follow this method, and even textbooks are written according to this method. However, this method gives good results if it is preceded by inductive method.

Thus we find that induction and deduction are complementary. So beginning should be made with the inductive method, which should then be followed by a sufficient drill work by deductive method. This promotes speed, accuracy and command of the subject. So understand mathematics inductively and apply deductively. Best results are given by induction-deduction combination.